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COSC 600

Programming Assignment 2

1.

```
<terminated> max_subseq [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe
Algorithm 1: Expected runtime  $O(n^3)$ 
n=5000. The actual runtime is 9.9870075s
n=10000. The actual runtime is 71.4431631s
n=50000, 100000 and 200000 were terminated due to the runtime.

Algorithm2: Expected runtime  $O(n^2)$ 
n=5000. The actual runtime is 0.0070021s
n=10000. The actual runtime is 0.0165926s
n=50000. The actual runtime is 0.3393313s
n=100000. The actual runtime is 1.3066959s
n=200000. The actual runtime is 5.3169852s

Algorithm3: Expected runtime  $O(n \log n)$ 
n=5000. The actual runtime is 0.0024934s
n=10000. The actual runtime is 6.225E-4s
n=50000. The actual runtime is 0.0050025s
n=100000. The actual runtime is 0.0110776s
n=200000. The actual runtime is 0.0067797s

Algorithm4: Expected runtime  $O(n)$ 
n=5000. The actual runtime is 7.09E-5s
n=10000. The actual runtime is 1.493E-4s
n=50000. The actual runtime is 7.521E-4s
n=100000. The actual runtime is 3.086E-4s
n=200000. The actual runtime is 9.549E-4s
```

2.

```
<terminated> josephus_problem [Java Application] C:\Program Files\Java\jdk-23\bin
How many soldiers?->
10
Type 10 soldiers name:
bob
tim
john
ash
bren
jerry
geoff
rick
alex
dani
Enter the postition->
3
1. john (3)
2. jerry (6)
3. alex (9)
4. tim (2)
5. geoff (7)
6. bob (1)
7. rick (8)
8. bren (5)
9. dani (10)
The survivor is: ash (4)
```

3.

```
<terminated> long_seq [Java Application] C:\Program Files\Java\jdk-23\bin\java
26 (1,4)
141 (2,5)
212 (2,4)
251 (3,3)
272 (2,3)
516 (1,3)
603 (1,2)
805 (2,2)
870 (3,2)
880 (4,1)
924 (3,0)
925 (2,0)
952 (2,1)
The max path is length: 13
```

```
<terminated> long_seq [Java Application] C:\Program Files\Java\jdk-23\bin\java
71 (10,6)
101 (9,7)
214 (8,7)
306 (9,8)
504 (10,9)
547 (10,10)
550 (11,9)
587 (10,8)
622 (11,8)
649 (12,8)
767 (13,9)
850 (13,10)
923 (12,11)
956 (11,10)
The max path is length: 14
```